

Sofiene Hamzaoui

Senior AI Systems Architect & Platform Engineer

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PROFESSIONAL SUMMARY

Senior Systems Architect and Platform Engineer with **20+ years** of hands-on experience designing, deploying, and optimising enterprise-grade AI infrastructure, local LLM inference clusters, high-performance backend pipelines, and security-centric crypto platforms. Expert in quantised GGUF model deployment (llama.cpp, CUDA/ROCm layer offloading, 128K context orchestration), custom Retrieval-Augmented Generation (RAG) vector engines, multi-agent autonomous systems, and high-throughput WebSocket architectures. Upwork **Top Rated Plus** — 100 % job-success rate across 50+ enterprise contracts.

PROFESSIONAL EXPERIENCE

Senior AI Systems Architect & Platform Consultant

Oct 2024 – Present

Adarma & Global Clients (Remote)

- Architected and deployed private local LLM inference engines using llama.cpp with custom CUDA/Vulkan offloading wrappers, handling context windows up to 128 K tokens.
- Designed self-hosted RAG architectures utilising ChromaDB 1.5 and sentence-transformer embeddings (all-MiniLM-L6-v2, nomic-embed) to securely index large-scale codebase semantics.
- Built multi-agent workflow runners mapping filesystems, auditing code schemas, and executing autonomous telemetry feedback loops via MCP protocol.
- Configured enterprise cloud inference using AWS Bedrock (us-east-1) with IAM-scoped policies, multi-provider fallback, and token-consumption auditing.
- Supervised secure deployment of OpenAI-compatible REST endpoints serving quantised GGUF models (Q4_K_M, IQ4_XS, Q8_K_P) to internal development teams.

Senior Systems Engineer & Backend Developer

Dec 2020 – Sep 2024

IronNet & Global Enterprises (Remote)

- Engineered high-throughput async telemetry pipelines using FastAPI, Celery, and Redis, reducing mean processing latency by 35 %.
- Developed multi-tenant dashboards deployed on Cloudflare Workers CDN serving global regions with sub-450 ms data-sync SLAs.
- Supervised migration, replication, and indexing for high-volume PostgreSQL clusters with zero service interruption.
- Built real-time monitoring environments using Prometheus & Grafana with policy-triggered alerting on network anomalies.
- Containerised development stacks using Docker Compose and Kubernetes, achieving repeatable CI/CD pipelines.

Full Stack & Infrastructure Engineer

Feb 2018 – Dec 2020

Rubica & SMB Portfolios (Remote)

- Designed and maintained scalable platforms for 20+ corporate and SMB clients, deploying secure backends and responsive UI portals.
- Hardened REST APIs with Zero-Trust access control, JWT authentication, and AES-256 state encryption.
- Automated infrastructure provisioning and containerisation using Docker, Docker Compose, and Bash workflows.
- Engineered document-ingestion tools parsing PDF, DOCX, and XLSX into structured database schemas.

Senior AI, Cybersecurity & Full-Stack Consultant

Apr 2017 – Present

Upwork — Top Rated Plus (100 % Job Success Rate)

- Successfully delivered 50+ projects for enterprise clients across North America, Europe, and the Middle East.
- Delivered custom APIs, complex database schemas, real-time WebSocket integrations, and vector-search pipelines.
- Maintained 100 % Client Satisfaction with continuous repeat business and long-term retainers.

FLAGSHIP PROJECTS

◆ **VaultGuard — Institutional P2P Crypto Settlement Desk**

TypeScript · Cloudflare Workers · Supabase · React · WebAuthn · ZKP

Production-deployed institutional OTC settlement engine running across Cloudflare's 300+ city edge network. Implements Zero-Knowledge Proof progressive identity vaults, passwordless WebAuthn/FIDO2 biometrics, multi-sig custody, and threat-intelligence ingestion (VirusTotal, AbuseIPDB, Shodan). 137 edge API endpoints live at vaultguard.eu.org.

◆ **Phalanx X — Local LLM OpenAI-Compatible API Server**

Python · llama.cpp · CUDA · GGUF · FastAPI

High-performance local inference server providing full OpenAI-compatible /v1/chat/completions endpoints. Automates CUDA/Vulkan GPU-layer offloading, supports MoE architectures (Qwen3.6 35B, 256 experts), 128 K context, and zero-latency streaming with no external dependencies.

◆ **AI-Integrated C2 Server — Autonomous Agent Command Loop**

Python · FastAPI · WebSockets · Redis · Docker · Ollama

Next-generation command & control dashboard integrating local LLM orchestration to direct autonomous scanning, environment mapping, and CVE lookup operations. Redis + PostgreSQL telemetry persistence. Multi-agent session management via FastAPI WebSockets with real-time feedback loops.

◆ **UK Insolvency Monitor — Enterprise Credit Risk Tracker**

Python · FastAPI · Celery · Redis · React · PostgreSQL

Real-time enterprise risk application ingesting UK public insolvency filings daily. Identifies credit-score adjustments ≥5 %, formats automated multi-sheet Excel reports via Celery async pipelines, and serves a multi-tenant dashboard with client-scoped filters.

◆ **Mythos Learner — Autonomous Repository Indexing Agent**

Python · Ollama · ChromaDB · sentence-transformers

Self-enhancing AI agent that scans project directories, builds unified semantic indexes from SKILL.md files, and auto-learns from 102+ capability specifications. Interfaces with local Qwen and Llama models for dynamic multi-step planning.

CORE COMPETENCIES

AI & LLM Engineering

Quantised Local Models (GGUF) · llama.cpp & CUDA Offloading · RAG & Vector Search · Multi-Agent Orchestration · Context Window Architecture · Fine-Tuning · MCP Protocol

Systems & Platform

FastAPI · Python · Node.js · WebSockets · Cloudflare Workers Edge API · PostgreSQL · Redis · Supabase · Celery · Docker · Kubernetes

Security & Cryptography

Zero-Knowledge Proofs (ZKP) · WebAuthn & FIDO2 · Zero-Trust API Hardening · Threat Intelligence & OSINT · BTCPay Server · AES-256 Encryption

Frontend & Tooling

React · TypeScript · Three.js · Tailwind CSS · Git · Linux · Bash · Aider · Continue IDE · Claude CLI · Open Interpreter

CERTIFICATIONS

- ◆ HuggingFace — AI Agents Course
- ◆ IBM — System & Network Security
- ◆ HuggingFace / DeepLearning.AI — Fundamentals of LLMs
- ◆ IBM — Cloud Security Architecture
- ◆ Microsoft / freeCodeCamp — Foundational C# with Microsoft
- ◆ IBM — GRC & Data Privacy
- ◆ IBM — Incident Response & System Forensics
- ◆ freeCodeCamp — Scientific Computing with Python
- ◆ IBM — Security Operations & Management
- ◆ freeCodeCamp — Frontend Development Libraries V8
- ◆ IBM — Vulnerability Management
- ◆ IELTS General Training — Band 5.5 / B2 (British Council)

LANGUAGES

Arabic	English	French	Russian	German
Native	Fluent (IELTS B2)	Fluent	Good	Basic

